<u>Meerut Institute OF Engineering and Technology, Meerut</u>

Session 2023-24

Mini project report

On

"Sorting Visualizer"

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING (DS)

Submitted to – Mrs. Himanshi Aggarwal

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DS)

Submitted by –

Student Name- 1. Disha Bansal (2100681540017)

2. Shagun Chauhan (2100681540042)

3. Shradha (2100681540044)

5th SEMESTER

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DS)
MEERUT INSTITUTE OF ENGINEERING AND TECHNOLOGY, MEERUT

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DECLARATION

We hereby declare that the project entitled - "**SORTING VISUALIZER**", which is being submitted as Mini Project in **Department of Computer Science and engineering (DS)** to Meerut Institute of Engineering and Technology, Meerut (U.P.) is an authentic record of our genuine work done under the guidance of Prof. "**Mrs. Himanshi Aggarwal**" of **Computer Science and Engineering (DS)**

Meerut Institute of Engineering and Technology, Meerut.

Name of Student – DISHA BANSAL

(2100681540017)

SHAGUN CHAUHAN

(2100681540042)

SHRADHA

(2100681540044)

Date: 14/12/2023

Place: - MIET , MEERUT

CERTIFICATE

This is to certify that mini project report entitled "SORTING VISUALIZER" submitted by "DISHA BANSAL, SHAGUN CHAUHAN and SHRADHA" has been carried out under the guidance of Prof. "**MRS.HIMANSHI AGGARWAL**" of Computer Science and Engineering (DS), Meerut Institute of Engineering and Technology, Meerut. This project report is approved for Mini Project (KCN 554) in 5th semester in "**Department of Computer Science And Engineering (DS)**" from Meerut Institute of Engineering and Technology, Meerut.

Internal Examiner Date:

ACKNOWLEDGEMENT

I express my sincere indebtedness towards our guide Prof., "**Mrs. Himanshi Aggarwal**" of Computer Science and Engineering (DS), Meerut Institute of Engineering and Technology, Meerut for his valuable suggestion, guidance and supervision throughout the work. Without his king patronage and guidance the project would not have taken shape. I would also like to express my gratitude and sincere regards for his kind approval of the project. Time to time counseling and advises.

I would also like to thank to our HOD Dr. (Prof.) "**Mr. Rohit Aggarwal"** Department of Computer Science and engineering, Meerut Institute of Engineering and Technology, Meerut for his expert advice and counseling from time to time.

I owe sincere thanks to all the faculty members in the department of Data Science for their kind guidance and encouragement time to time.

Date: **Student Name**: 1. DISHA BANSAL

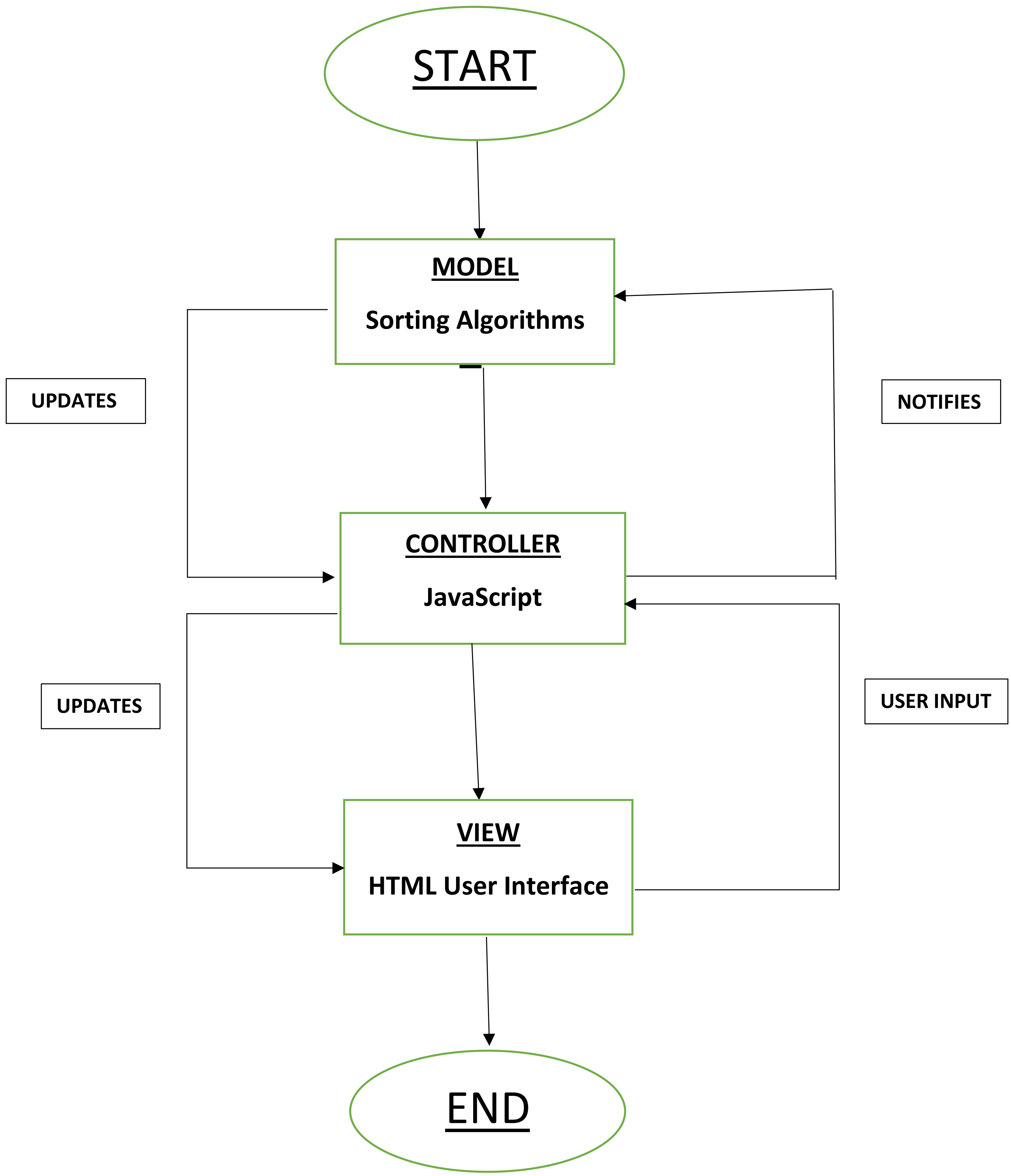
2. SHRADHA
3. SHAGUN CHAUHAN

CHAPTER 1
INTRODUCTION

1. We have created a visualizer project in which when user land on page for 1st time he/she will see a page where they have an options to Change an array. There is also a different buttons to Change the techniques i.e. Selection Sorting , Insertion Sort , Merge Sorting , Heap Sorting etc.

2. User have to click on that button and after clicking on the button they will be see an animation of an array.
3. For creating this project we uses HTML, CSS , JS.

CHAPTER 2 SYSTEM DESIGN

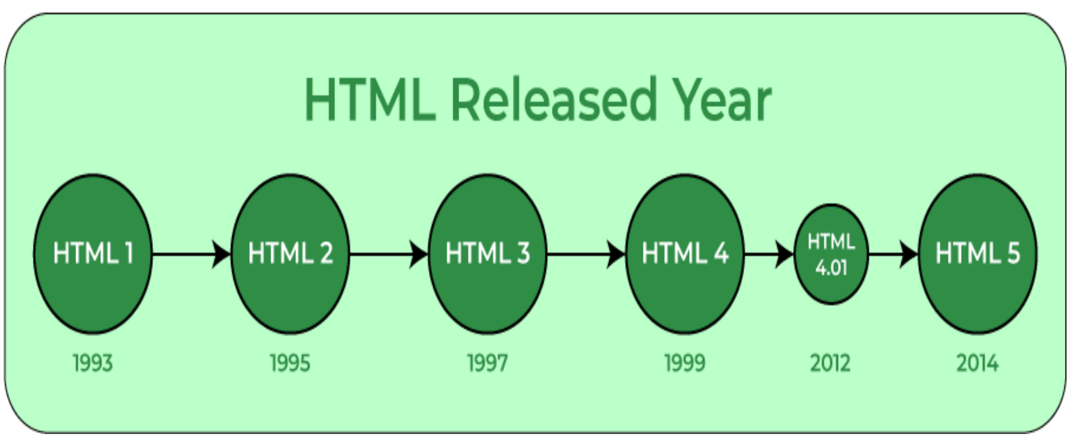


CHAPTER 3

TECHNOLOGY BUCKET

3. Description Html

HTML-HTML stands for Hyper Text Markup Language. It is used to design web pages using a markup language. HTML is the combination of hypertext and markup language. Hypertext defines the link between web pages. A markup language is used to define the text document within the tag which defines the structure web pages. This language is used to annotate (make notes for the computer) text so that a machine can understand it and manipulate text accordingly. Most markup languages (eg. HTML) are human-readable. The languages uses tags to define what manipulate has to be done .



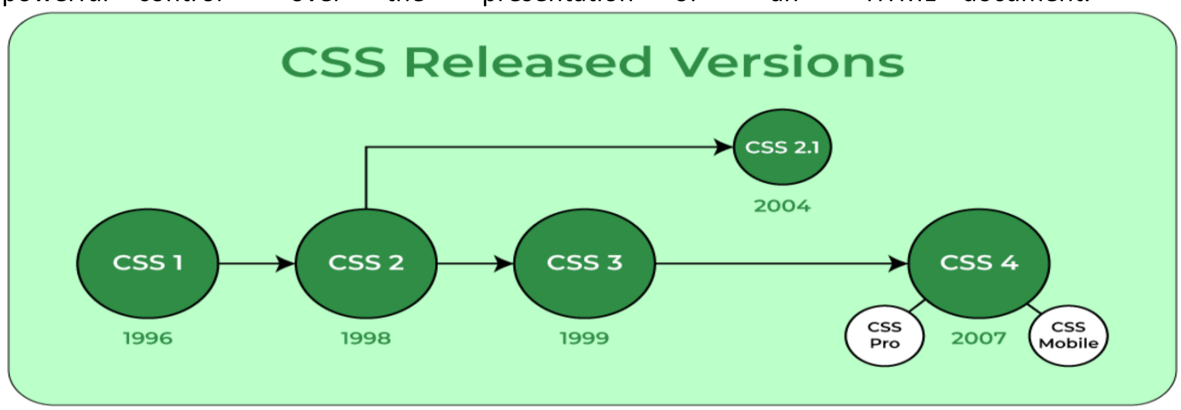
ELEMENTS AND TAGS- HTML uses predefined tags and elements which tell the browser how to properly display the content. Remember to include closing tags. If omitted, the browser applies the effect of the opening tag until the end of the page.

HTML PAGE STRUCTURE- The basic structure of an
HTML page is laid out below. It contains the essential buildingblock elements (i.e doctype declaration, HTML, head, title, and body elements) upon which all web pages are crated.

4. Description Of CSS

CSS - Cascading Style Sheets, fondly referred to an CSS, is a simply designed language intended to simplify the process of making web pages presentable. CSS allows you to apply styles to web pages. More importantly, CSS enables you to do this independent of the HTML that makes up each web page. It describes how a web page should look: it prescribes colors, fonts, spacing, and much more. In short, you can make your website look however you want. CSS lets developers and designers define how it behaves, including how elements are positioned in the browser.

While HTML uses tags, css use rulesets. CSS is easy to learn and understand, but it provides powerful control over the presentation of an HTML document.



CSS SYNTAX- CSS comprises style rule that are interpreted by the browser and then applied to the corresponding elements in your document.

A style rule set consists of a selector and declaration block.

Selector – h1

Declaration – {color: blue; font size:12px} the selector points to the HTML element you want to style.

The declaration block contains one or more declarations separated by semicolons.

Each declaration includes a CSS property name and a value, separated by a colon.

<u>For example</u>- color is property and blue is value.

Font-size is property and 12px is value.

CSS declaration always ends with a semicolon, and declaration block are surrounded by curly braces.

5. DESCRIPTION OF JAVASCRIPT

JAVASCRIPT – JAVASCRIPT is the world's most popular lightweight, interpreted compiled programming language. It is also known as scripting language for web pages. It can be used for client-side as well as server-side developments.

JAVASCRIPT can be added to your HTML file in two ways:

- INTERNAL JAVASCRIPT – We can add JAVACSRIPT code directly to your HTML file by writing the code inside the <script> tag. The <script> tag can either be placed inside the <head> or the <body> tag according to the requirement.
- EXTERNAL JAVASCRIPT FILE – We can create a file with.js extension and paste the JAVASCRIPT code inside it. After creating the file, add this file in <script src= "file_name.js"> tag inside <head> tag of the HTML file.

SYNTAX- <script>

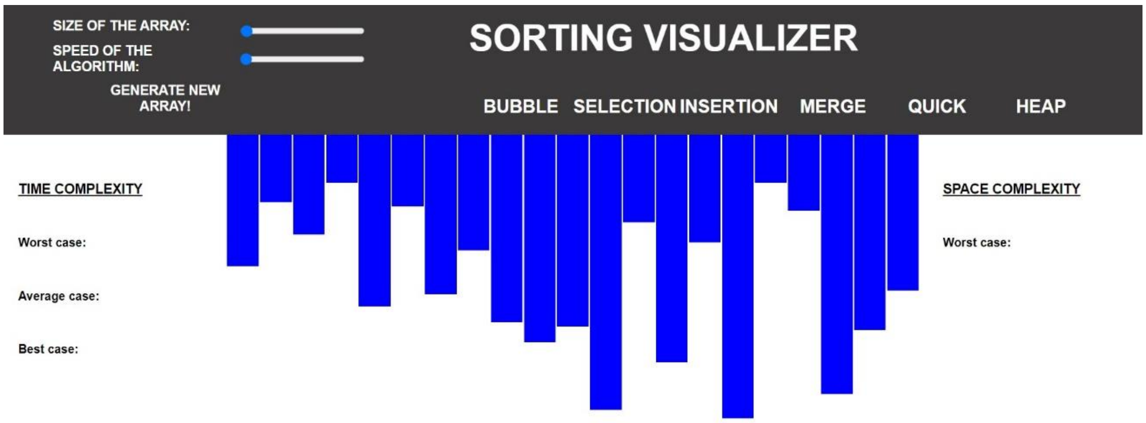
// JAVASCRIPT code

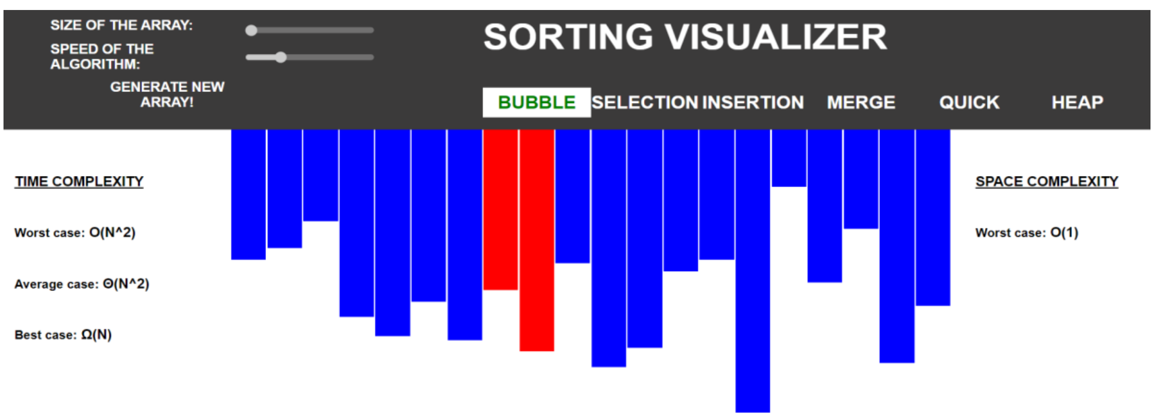
</script>

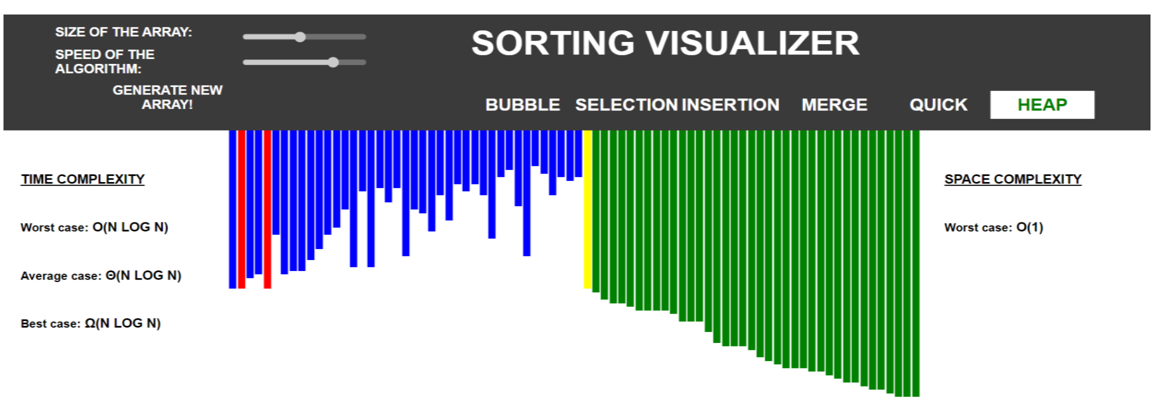
Example – its is the basic example of JAVASCRIPT code embedded in HTML file.

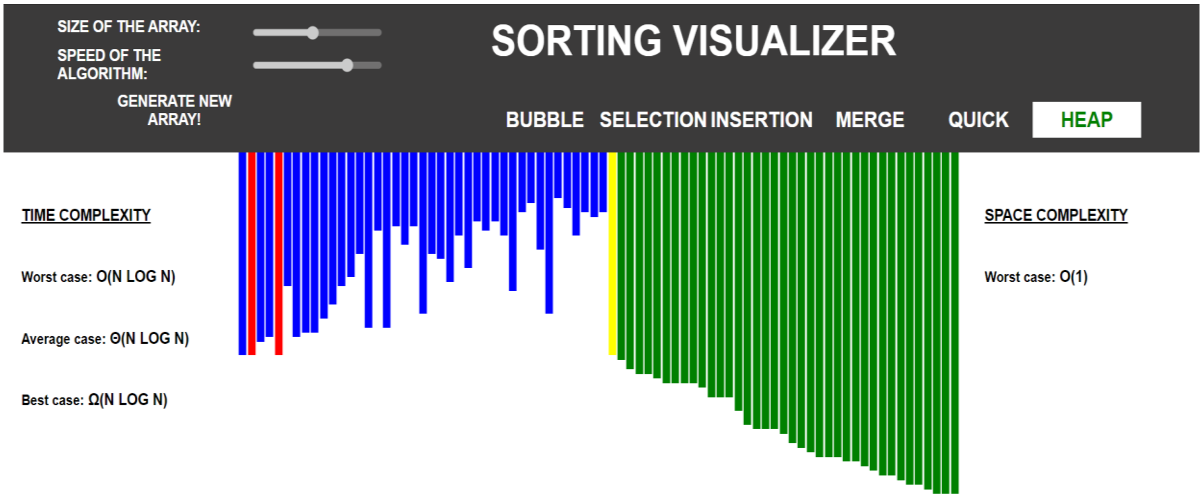
CHAPTER 4

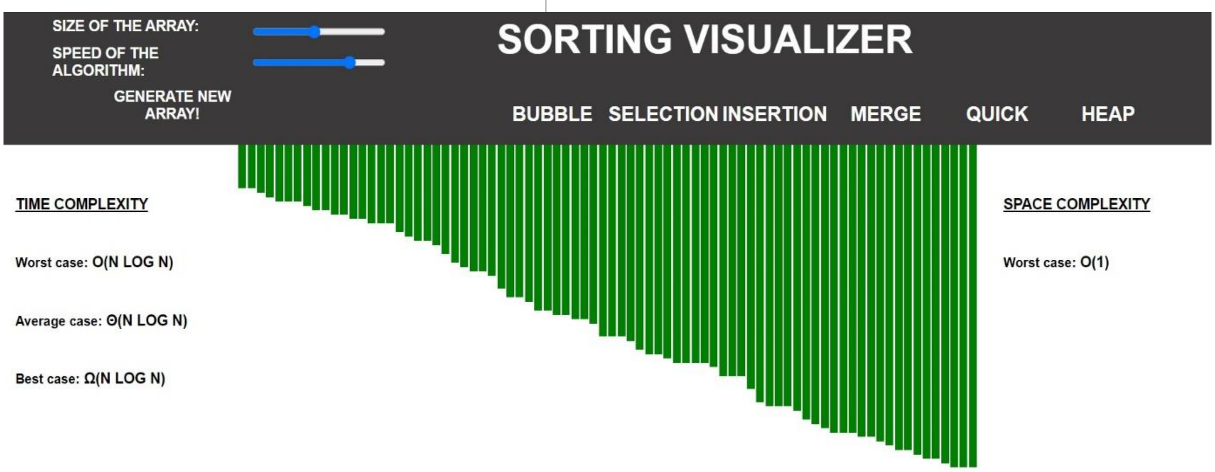
SOME SCREENS

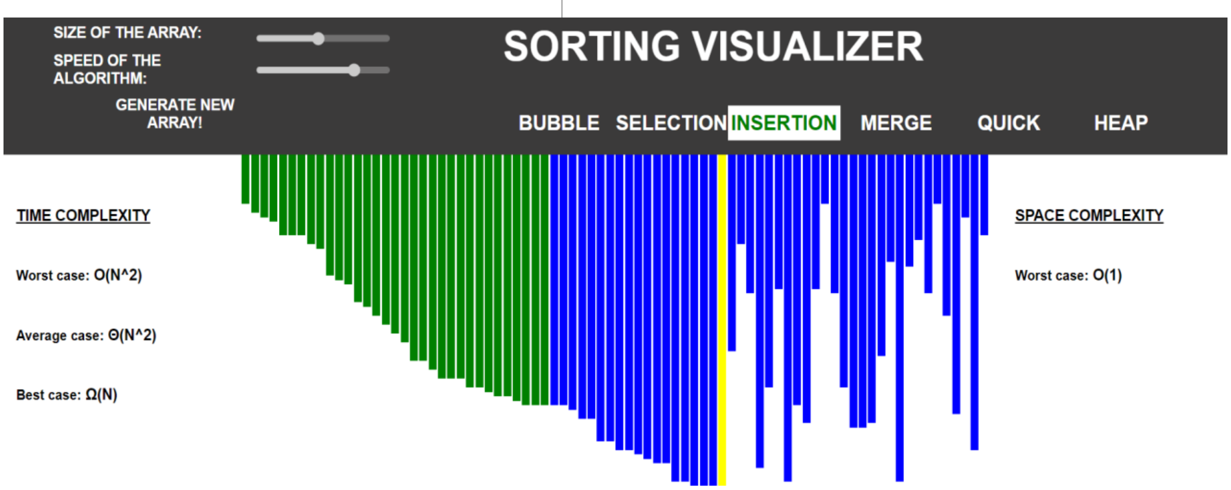







SIZE OF THE ARRAY:
SPEED OF THE ALGORITHM:
GENERATE NEW ARRAY!
SORTING VISUALIZER
BUBBLE
SELECTION
INSERTION
MERGE
QUICK
HEAP
TIME COMPLEXITY
Worst case: O(N LOG N)
Average case: Θ(N LOG N)
Best case: Ω(N LOG N)
SPACE COMPLEXITY
Worst case: O(1)


SIZE OF THE ARRAY:
SPEED OF THE ALGORITHM:
GENERATE NEW ARRAY!
SORTING VISUALIZER
BUBBLE
SELECTION
INSERTION
MERGE
QUICK
HEAP
TIME COMPLEXITY
Worst case: O(N LOG N)
Average case: Θ(N LOG N)
Best case: Ω(N LOG N)
SPACE COMPLEXITY
Worst case: O(1)


SIZE OF THE ARRAY:
SPEED OF THE ALGORITHM:
GENERATE NEW ARRAY!
SORTING VISUALIZER
BUBBLE
SELECTION
INSERTION
MERGE
QUICK
HEAP
TIME COMPLEXITY
Worst case: O(N^2)
Average case: Θ(N^2)
Best case: Ω(N)
SPACE COMPLEXITY
Worst case: O(1)

Appendices

Implementation Code

The Main Page

Html

index.html

index.html > html > head > link

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sorting Visualizer</title>

    <link rel="stylesheet" href="./css/style.css">
</head>
<body>

    <header>
        <nav>
            <div class="array-inputs">
                <p>Size of the array:</p>
                <input id="a_size" type="range" min=20 max=150 step=1 value=80>
                <p>Speed of the algorithm:</p>
                <input id="a_speed" type="range" min=1 max=5 step=1 value=4>
                <button id="a_generate">Generate New Array!</button>
            </div>

            <div class="header_right">
                <p class="nav-heading">Sorting visualizer</p>

                <div class="algos">
                    <button >Bubble</button>
                    <button >Selection</button>
                    <button >Insertion</button>
                    <button >Merge</button>
                    <button >Quick</button>
                    <button style="margin-right: 0px;">Heap</button>
                </div>
            </div>

        </nav>
    </header>
```

```html
        </nav>
</header>
    <section>

        <div id="Info_Cont1">

            <h3>TIME COMPLEXITY</h3>

            <div class="Complexity_Containers" id="Time_Complexity_Types_Container">

                <div class="Pair_Definitions">
                    <p class="Sub_Heading">Worst case:</p>
                    <p id="Time_Worst"></p>
                </div>

                <div class="Pair_Definitions">
                    <p class="Sub_Heading">Average case:</p>
                    <p id="Time_Average"></p>
                </div>

                <div class="Pair_Definitions">
                    <p class="Sub_Heading">Best case:</p>
                    <p id="Time_Best"></p>
                </div>

            </div>

        </div>

        <div id="array_container">

        </div>

        <div id="Info_Cont2">
```

```html
        <div id="Info_Cont2">

            <h3>SPACE COMPLEXITY</h3>

            <div class="Complexity_Containers" id="Space_Complexity_Types_Container">

                <div class="Pair_Definitions">
                    <p class="Sub_Heading">Worst case:</p>
                    <p id="Space_Worst"></p>
                </div>

            </div>

        </div>

    </section>

    <footer>

    </footer>

    <script src="./scripts/main.js"></script>   <!--This should be at the end of body and should be loaded before sorts.js-->
    <script src="./scripts/visualizations.js"></script>  <!--This should be loaded after main.js-->

    <script src="./scripts/bubble_sort.js"></script>
    <script src="./scripts/selection_sort.js"></script>
    <script src="./scripts/insertion_sort.js"></script>
    <script src="./scripts/merge_sort.js"></script>
    <script src="./scripts/quick_sort.js"></script>
    <script src="./scripts/heap_sort.js"></script>
</body>


</html>
```

Sorting Algorithm

Bubble Sort

scripts > JS bubble_sort.js > ...

```
function Bubble()
{
    //Setting Time complexities
    document.getElementById("Time_Worst").innerText="O(N^2)";
    document.getElementById("Time_Average").innerText="Θ(N^2)";
    document.getElementById("Time_Best").innerText="Ω(N)";

    //Setting Space complexity
    document.getElementById("Space_Worst").innerText="O(1)";

    c_delay=0;

    for(var i=0;i<array_size-1;i++)
    {
        for(var j=0;j<array_size-i-1;j++)
        {
            div_update(divs[j],div_sizes[j],"yellow");//Color update

            if(div_sizes[j]>div_sizes[j+1])
            {
                div_update(divs[j],div_sizes[j], "red");//Color update
                div_update(divs[j+1],div_sizes[j+1], "red");//Color update

                var temp=div_sizes[j];
                div_sizes[j]=div_sizes[j+1];
                div_sizes[j+1]=temp;

                div_update(divs[j],div_sizes[j], "red");//Height update
                div_update(divs[j+1],div_sizes[j+1], "red");//Height update
            }
            div_update(divs[j],div_sizes[j], "blue");//Color updat
        }
        div_update(divs[j],div_sizes[j], "green");//Color update
    }
    div_update(divs[0],div_sizes[0], "green");//Color update
```

Insertion Sort

index.html ● bubble_sort.js ● insertion_sort.js ●

scripts > insertion_sort.js > ...

```javascript

function Insertion()
{
    //Setting Time complexities
    document.getElementById("Time_Worst").innerText="O(N^2)";
    document.getElementById("Time_Average").innerText="Θ(N^2)";
    document.getElementById("Time_Best").innerText="Ω(N)";

    //Setting Space complexity
    document.getElementById("Space_Worst").innerText="O(1)";

    c_delay=0;

    for(var j=0;j<array_size;j++)
    {
        div_update(divs[j],div_sizes[j],"yellow");//Color update

        var key= div_sizes[j];
        var i=j-1;
        while(i>=0 && div_sizes[i]>key)
        {
            div_update(divs[i],div_sizes[i],"red");//Color update
            div_update(divs[i+1],div_sizes[i+1],"red");//Color update

            div_sizes[i+1]=div_sizes[i];

            div_update(divs[i],div_sizes[i],"red");//Height update
            div_update(divs[i+1],div_sizes[i+1],"red");//Height update

            div_update(divs[i],div_sizes[i],"blue");//Color update
            if(i==(j-1))
            {
                div_update(divs[i+1],div_sizes[i+1],"yellow");//Color update
            }
            else
            {
                div_update(divs[i+1],div_sizes[i+1],"blue");//Color update
```

```javascript
            else
            {
                div_update(divs[i+1],div_sizes[i+1],"blue");//Color update
            }
            i-=1;
        }
        div_sizes[i+1]=key;

        for(var t=0;t<j;t++)
        {
            div_update(divs[t],div_sizes[t],"green");//Color update
        }
    }
    div_update(divs[j-1],div_sizes[j-1],"green");//Color update

    enable_buttons();
}
```